

up must be on the same line of action (dotted line is vertical)

B1

- 4 (a) The gravitational field strength at a point in a gravitational field is defined as the **gravitational force exerted per unit mass** acting on a small mass placed at that point. B1

- (b) (i) point X is on the line and closer to Moon B1

(ii) Gravitational field strength at A = $\frac{G(5.97 \times 10^{24})}{(6.37 \times 10^6)^2} - \frac{G(7.34 \times 10^{22})}{(3.84 \times 10^8 - 6.37 \times 10^6)^2} = 9.81 \text{ m s}^{-2}$

Gravitational field strength at B = $\frac{G(7.34 \times 10^{22})}{(1.74 \times 10^6)^2} - \frac{G(5.97 \times 10^{24})}{(3.84 \times 10^8 - 1.74 \times 10^6)^2} = 1.61 \text{ m s}^{-2}$

Marks awarded as follow for both calculations:

- correct equations that include contributions by both Earth and Moon M1

- correct substitutions (e.g. unit conversion, etc) M1

- correct final values A1

- (iii) Shape of graph, one positive one negative, cut x-axis closer to B. B1

5 (a) (i) Length of path 1 = $1\,000 - 0\,300 = 0\,700\text{ m}$